

~~Assessment~~

Assessment Tool-2

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CSA1613 - Data Warehousing and Data Mining

Q1) Define Data mining and KDD process.

AS Definition of Data mining:-

Data mining is the process of extracting useful, hidden and meaningful patterns or knowledge from large volumes of data using either statistical.

KDD:-

- 1) Data Selection
- 2) Data Cleaning
- 3) Data Integration
- 4) Data Transformation



⑥ Data Cleaning:-

Data Cleaning is the process of detecting and correcting inaccurate, incomplete, duplicate or inconsistent data.

⇒ Ex. Age 20, 22, -, 24

$$\text{mean} = (20 + 22 + 24) / 3 = 22$$

missing value = 22
#

The Two common data Quality Issues

→ Missing values

→ noisy data

Qd (a) Data Integration:-

Data integration is the process of combining data from multiple heterogeneous sources into a single consistent data set.

→ Data redundancy.

→ Data inconsistency.

* Handling Redundancy:-

→ Remove duplicate records.

→ Use correlation analysis.

(b) Covariance Analysis:-

Covariance analysis measures how two variables change together.

* Formula

$$Cov(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

* Sample Covariance:-

$$Cov(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

a3) (a) Stock Analysis:-

As Given:-

<u>A</u>	<u>B</u>
2	5
3	8
5	10
4	11
6	14

As Stock A increases, Stock B also generally increases.

⇒ The Correlation is ~~sp~~ positive, so the two stocks rise together.

(b)

Data Sources



Data warehouse



Data Cleaning & Integration



Data base server.



Data mining Engine



Pattern Evaluation module.



User / Decision maker.

Q4) a) Covariance

Given

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{6, 3, 7, 5, 6\}$$

mean

$$\bar{X} = \frac{31}{5} = 6.2$$

$$\bar{Y} = \frac{29}{5} = 5.8$$

$$= \text{Sum} = -1.80$$

$$= \text{Cov} = \frac{-1.80}{5} = -0.36$$

$$\text{Answer:- Covariance} = -0.36$$

4)

Major steps:-

Data cleaning

Data Integration

Data Transformation

Data Reduction

→ This sequence improves data quality removes inconsistencies and prepares the data mining.

5) a) Equal - Frequency (equi-depth) Binning.

Data:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40,
45, 46, 52, 70

Total values = 27.

Bin 1 :-

13, 15, 16, 16, 19, 20, 20, 21, 22.

mean = 18

Smoothed values

18, 18, 18, 18, 18, 18, 18, 18.

Bin 2 :-

22, 25, 25, 25, 30, 33, 33, 35

mean = 28

Smoothed values

28 = 28, 28, 28, 28, 28, 28, 28, 28.

b) Data Discretization

Data discretization Converts continuous numerical values
into intervals (bins).

= Equal-width Binning

= Data 10-100

width = 30

bins

→ 10-40

→ 40-70

→ 70-100

Advantage:

→ simple

→ Easy to implement.

Disadvantages:

→ sensitive to outliers:

